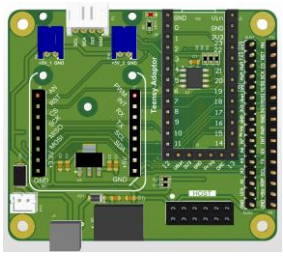
MAKE THINGS HAPPY !

Teensy Adaptor

mikroBUS™ Adaptor for Teensy Modules



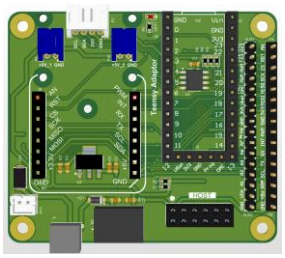
Teensy Adaptor module (rev. 2.21) is designed for installing **Teensy** modules (**Teensy 4.0**, **Teensy 3.2**, **Teensy LC**) and **mikroBUS™** modules, as well as for connecting **IoTextra** series mezzanine modules via a cable.

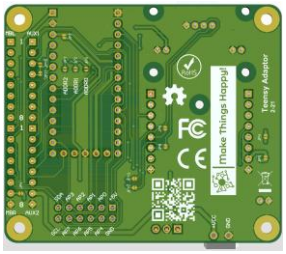
The module has an on-board **LM75B** temperature sensor connected to the **I²C** bus.

The module's input power is non-isolated, ranging from 9VDC to 36VDC. This power is supplied via a 2-pin terminal block with a 3.5mm pitch.

The module has the following slots:

- **TEENSY** (**M2** slot, **J8**, **J2** and **J3** connectors) for installing **Teensy** modules
- **HOST** for connecting **IoTextra** series mezzanine modules via a cable
- **M1** slot (**J6** and **J7** connectors) compatible with **mikroBUS™**





The pinout of the **HOST** connector corresponds to the **HOST-P12** connector of the **IoTbase** series modules, so any **IoTextra** series mezzanine can be connected to the **Teensy Adaptor** via a 12-pin cable.
The **I²C** interface is often used when designing devices and instruments based on the **Teensy Adaptor**, so the module has an **I²C** connector (**P2**) for connecting external devices and sensors. By default, pull-up resistors for **I²C** are connected, but they can be disconnected.
The signals of the **TEENSY** slot that are not used by the **mikroBUS™** slot and the **HOST** connector are routed to the **AUX1** and **AUX2** auxiliary connectors.
The input voltage of 9VDC to 36VDC is converted to 5VDC by a converter. The maximum current is 1A. The module also has a linear regulator from 5VDC to 3.3VDC. The 5VDC voltage can be distributed to other modules via two connectors (**5V-1** and **5V-2**), and the input voltage via the **VIN** connector.
Since the modules installed in the **TEENSY** slot and in the **M1** slot can themselves generate 3.3VDC, the **Teensy Adaptor** provides the ability to flexibly manage these connections using jumpers on its bottom side.
The module size is 65×56 mm. The module format corresponds to the popular **Raspberry Pi 3A+** board format, which greatly simplifies its use.
Primary applications of the module:

- Industry IoT
- Distributed Data Acquisition Systems
- Motor control
- Heating, Ventilation, & Air Condition Monitoring (HVAC)
- Smart sensors
- PLC
- Smart home
- Consumer electronics

QUICK START

The **Teensy Adaptor** is most often used with one or another **IoTextra** series mezzanine, as together with a connected mezzanine and a **Teensy** module it is easy to use as a standalone device. The **Teensy** module is installed in the **TEENSY** slot, and the **IoTextra** mezzanine is connected to the **HOST** connector via a 12-pin cable (2×6, 2.54mm pitch, pin-to-pin connection 1–1 … 12–12).

Warning! You must pay attention to the orientation of the modules installed in the TEENSY slot. The USB connector of the installed module must face the edge of the board where the 5V-1, 5V-2 and I²C (P2) connectors are located. Incorrect installation can destroy the Teensy Adaptor and the Teensy module.

The following **IoTextra** series mezzanines can be connected to the **Teensy Adaptor** via the **HOST** connector:

- **IoTextra Input**
- **IoTextra Relay2**
- **IoTextra SSR Small**
- **IoTextra MOSFET2**
- **IoTextra Octal**, **IoTextra Octal2**, **IoTextra Octal3** and **IoTextra Octal4**
- **IoTextra Analog** and **IoTextra Analog3**
- **IoTextra Combo**

The current list of **IoTextra** series mezzanines is available on the website **http://www.makethingshappy.io**.

The **M1** slot allows the use of more than 2000 **MikroElektronika** **Click®** modules with the **mikroBUS™** bus:



To install **Click®** modules into the **mikroBUS™** slot, you don't need to do anything with the jumpers on the **Teensy Adaptor** (you can use the default settings for all jumpers on the module).

Numerous sensors, peripherals, and modules can also be easily connected to the **Teensy Adaptor** via the **I²C** connector (for example, **Grove** modules from Seeed Studio, using an appropriate cable):



DESCRIPTION

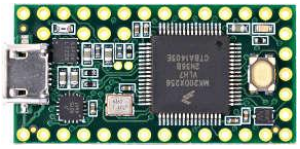
The **Teensy Adaptor** has:

- **J8**, **J2** and **J3** connectors of the **TEENSY** slot (**M2**)
- **J6** and **J7** connectors of the **M1** slot for the **mikroBUS™** bus
- **HOST** connector for connecting to **IoTextra** mezzanines
- **AUX1** and **AUX2** auxiliary connectors
- **MBL** (**P1**) and **MBR** (**P8**) connectors with the **mikroBUS™** bus signals (not installed, but can be self-installed by the user)
- **I²C** connector (**P2**) for the **I²C** bus
- **LM75B** temperature sensor
- **PWR** power connector (**J1**) for 9-36VDC power
- **VIN**, **5V-1** and **5V-2** connectors for power distribution

TEENSY slot. This slot consists of two 14-pin connectors (**J8** and **J2**) and a 5-pin connector (**J3**) for the end row of pins of the **Teensy** module. The following **Teensy** modules can be installed into this slot of the **Teensy Adaptor**:



Teensy 4.0



Teensy 3.2



Teensy LC



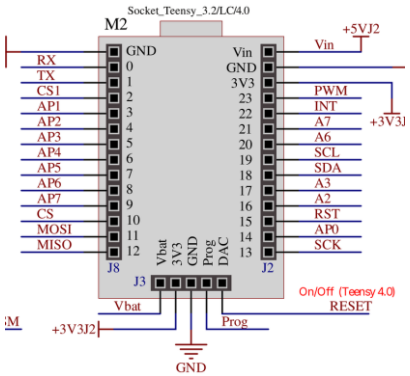
Teensy 3.1 (previously produced)

Key features of the **Teensy** modules (according to **PJRC**):

Feature	Teensy LC	Teensy 3.2	Teensy 4.0
MCU	MKL26Z64VFT4	MK20DX256VLH7	IMXRT1062DVL6
Core	Cortex-M0+	Cortex-M4	Cortex-M7
Rated speed, MHz	48	72	600
Flash memory, KB	62	256	1984
RAM, KB	8	64	1024
EEPROM, bytes	128 (emu)	2048	1080 (emu)
DMA channels	4	16	32
Digital I/O pins	27	34	40
Voltage output	3.3V / 5V	3.3V	3.3V
Current output	5mA / 20mA	10mA	10mA
Voltage input	3.3V only	5V tolerant	3.3V only
Analog input pins	13	21	14
ADC usable resolution, bits	12	13	10
Touch sensing pins	11	12	—
DAC pins / resolution, bits	1 / 12	1 / 12	—
PWM outputs	10	12	27
RTC	—	1	1
USB	1	1	2
Serial	3	3	7
SPI	2	1	2
I²C	2	2	3
CAN bus	—	1	3
Digital audio in / out	1 / 1	2 / 2	5 / 5
S/PDIF in / out	—	—	1 / 1

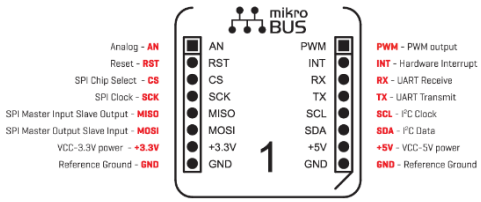
Attention! Teensy modules have the VIN and VUSB pads connected. If the Teensy module is connected to a computer via USB while the Teensy Adaptor is powered, cut the VIN-VUSB pad on the Teensy module or open jumper JP5.

Slot structure:

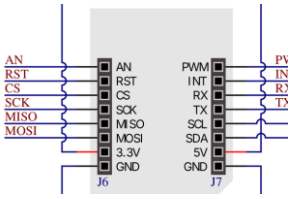


The 5th pin of the **J3** connector is **DAC** (A14) for **Teensy 3.2** and **On/Off** for **Teensy 4.0**. On the **Teensy Adaptor** this pin is connected to the **RESET** signal of the **AUX1** connector.

mikroBUS™. The module has an **M1** slot (connectors **J6** and **J7**) which complies with the **mikroBUS™** standard:



The **M1** slot structure on the **Teensy Adaptor**:



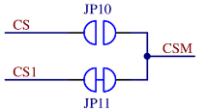
The **+3.3V** and **+5V** power of the **M1** slot can be disconnected from the module power using jumpers **JP6** and **JP4**, and the **SCL** and **SDA** signals can be disconnected from the common **I²C** bus using jumpers **JP8** and **JP9**.

Note. The AN signal of the M1 slot is not connected to the TEENSY slot (all TEENSY slot pins are used for other signals). The AN signal is available only on the MBL connector.

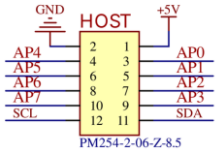
MBL and MBR Connectors. The **MBL** (**P1**) and **MBR** (**P8**) connectors (2.54mm, 1×8, not installed by default) duplicate the **mikroBUS™** bus signals. Thus, through these connectors it is easy to connect another module (or even several) with **mikroBUS™** slots, increasing the number of slots on the common bus. Connector pinout:

MBL pin	Signal	MBR pin	Signal
1	AN	1	PWM
2	RST	2	INT
3	CSM	3	RX
4	SCK	4	TX
5	MISO	5	SCL
6	MOSI	6	SDA
7	+3V3	7	+5V
8	GND	8	GND

The **CSM** chip select signal of the **MBL** connector is selected using jumpers **JP10** and **JP11**: **CS** (pin 10 of the **Teensy** module) or **CS1** (pin 2 of the **Teensy** module). By default, **CS1** is selected (JP10 is OPEN, JP11 is CLOSE). Only one of these jumpers can be closed at a time.

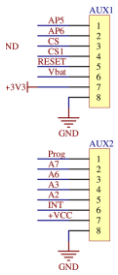


HOST Connector. This connector is for connecting application mezzanine modules (**IoTextra** series) to the **Teensy Adaptor** via a 12-pin cable. The connector type is a 2×6 pin socket with a 2.54mm pitch. Connector pinout:



The **AP0**-**AP7** signals are general-purpose signals of the mezzanine. Their correspondence to the pins of the **TEENSY** slot is given in the **SIGNAL MATCHING** section.

AUX1 and AUX2 Auxiliary Connectors. Almost all signals of the **TEENSY** slot that are not used by the **mikroBUS™** slot are routed to the **AUX1** and **AUX2** connectors (2.54mm, 1×8), which makes it possible to use almost all **Teensy** functions. The structure of these connectors is as follows:

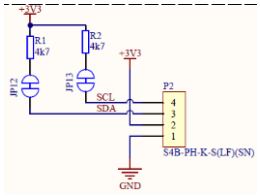


The **AUX2** connector pin 7 (**+VCC**) carries the input voltage of the module (9VDC to 36VDC).

Attention! In rev. 2.21 the top side silkscreen of the AUX1 connector shows the functions of the corresponding Teensy pins, and pin 6 of the AUX2 connector is labeled A1, while it carries the INT signal (Teensy pin 22). Use the pinout given in this document:

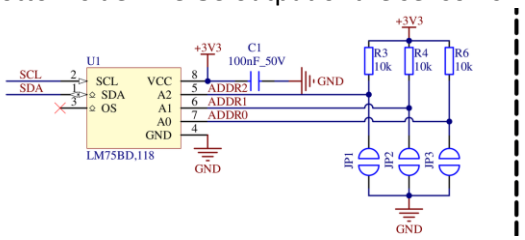
AUX1 pin	Top silkscreen	Actual signal	AUX2 pin	Top silkscreen	Actual signal
1	RX2 / RX3	AP5 (pin 7)	1	Prog	Prog
2	TX2 / TX3	AP6 (pin 8)	2	A7	A7 (pin 21)
3	PWM	CS (pin 10)	3	A6	A6 (pin 20)
4	PWM	CS1 (pin 2)	4	A3	A3 (pin 17)
5	PWM / DAC	RESET (DAC / On/Off)	5	A2	A2 (pin 16)
6	Vbat	Vbat	6	A1	INT (pin 22)
7	3V3	+3V3	7	VIN	+VCC
8	GND	GND	8	GND	GND

I²C Connector. The **I²C** connector (**P2**) is a 4-pin JST PH connector with a 2.0mm pitch (**S4B-PH-K-S**) and has the following structure:



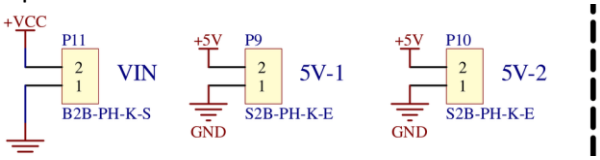
By default, 4.7 kOhm pull-up resistors are connected to the **SDA** and **SCL** signals (jumpers **JP12** and **JP13**).

Temperature Sensor. The module has a temperature sensor - an **LM75B** chip, connected via **I²C**. Temperature measurement accuracy: ±2°C for the temperature range from −25°C to +100°C, ±3°C for the temperature range from −55°C to +125°C. The **LM75B** chip can be replaced by the user with the **TMP1075** chip (maximum accuracy ±1°C).
The **I²C** address of the **LM75B** is 1001111x (0x4F) by default. This address can be changed by the user using jumpers **JP1**, **JP2** and **JP3** on the bottom side. The **OS** output of the sensor is not used.



POWER SUPPLY

The module's input power is non-isolated, ranging from 9VDC to 36VDC. This power is supplied via a 2-pin terminal block with a 3.5mm pitch (**PWR** connector, labeled **J1** on the module's schematic). Protection against incorrect input power polarity (Schottky diode **DSK34**) and a transient voltage suppressor (**SMAJ40CA**) are provided.
The input voltage of 9VDC to 36VDC is converted to 5VDC by a **K7805M-1000R3** converter. The maximum current is 1A.
The module also has a linear regulator from 5VDC to 3.3VDC (**AMS1117-3.3**). The total current consumed from the 5VDC (including the 3.3VDC consumers) must not exceed 1A.
The presence of 5VDC is indicated by a red **LED1**.
The 5VDC voltage can be distributed to other modules via the **5V-1** (**P9**) and **5V-2** (**P10**) connectors, and the input voltage (before the polarity protection diode) via the **VIN** (**P11**) connector. All these connectors are 2-pin JST PH connectors with a 2.0mm pitch:



By default, the 5VDC of the module is connected to the **Vin** pin of the **TEENSY** slot (jumper **JP5**), and the 3.3VDC is connected to the **3V3** pins of the **TEENSY** slot (jumper **JP7**). Since the **Teensy** module generates its own 3.3VDC, it is recommended to open jumper **JP7** if the 3.3VDC of the adaptor and of the **Teensy** module should not be connected together.

SIGNAL MATCHING

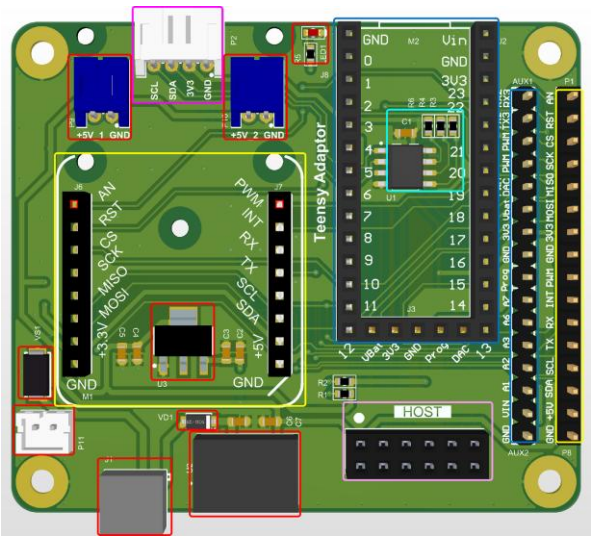
The **Teensy Adaptor** combines the **Teensy**, **mikroBUS™** and **HOST** (for communication with mezzanines) buses. The correspondence between the signals of the various buses in the **Teensy Adaptor** is shown in the following table (the **J8**, **J2** and **J3** columns contain the pin names of the **Teensy** module):

	Schema	TEENSY			mikroBUS (M1)		MBL / MBR	HOST	AUX1	AUX2
		J8	J2	J3	LEFT	RIGHT				
UART	RX	0				RX	MBR-3			
	TX	1				TX	MBR-4			
I2C	SDA		18			SDA[1]	MBR-6	11		
	SCL		19			SCL[1]	MBR-5	12		
SPI	SCK		13		SCK		MBL-4			
	MISO	12			MISO		MBL-5			
	MOSI	11			MOSI		MBL-6			
	CS	10			CS		MBL-3[2]		3	
	CS1	2					MBL-3[2]		4	
RST	RST		15		RST		MBL-2			
	RESET			DAC / On/Off					5	
PWM	PWM		23			PWM	MBR-1			
INT & AN	INT		22			INT	MBR-2			6
	AN				AN		MBL-1			
AP0-AP7	AP0		14					3		
	AP1	3						5		
	AP2	4						7		
	AP3	5						9		
	AP4	6						4		
	AP5	7						6	1	
	AP6	8						8	2	
	AP7	9						10		
OTHER	A2		16							5
	A3		17							4
	A6		20							3
	A7		21							2
	Vbat			Vbat					6	
	Prog			Prog						1

[1] The **SCL** and **SDA** signals of the **M1** slot are connected to the common **I²C** bus via jumpers **JP8** and **JP9** (closed by default).
[2] The **CSM** signal of the **MBL** connector is **CS** or **CS1**, selected by jumpers **JP10** and **JP11** (**CS1** by default).
The **RX** and **TX** signals of the **TEENSY** slot are connected directly to the **RX** and **TX** pins of the **mikroBUS™** slot.
The **AN** signal of the **mikroBUS™** slot is connected only to the **MBL** connector.

LAYOUT

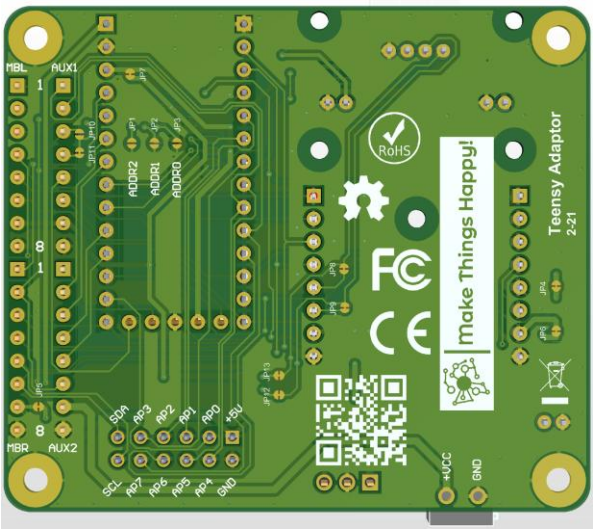
Below is the layout of the elements on the top side of the **Teensy Adaptor**:



In this picture:

- **Red** highlights power-related elements (external PWR power connector J1, VIN, 5V-1 and 5V-2 connectors, VCC -> 5VDC converter, 5VDC -> 3.3VDC regulator, protection elements, and LED for 5VDC power indication)
- **Blue** highlights the TEENSY slot connectors (J8, J2 and J3) and the AUX1 and AUX2 auxiliary connectors
- **Yellow** highlights the mikroBUS™ slot M1 (J6 and J7) and the MBR and MBL connectors
- **Pink** highlights the HOST connector for communication with mezzanines
- **Purple** highlights the I²C connector for connecting peripherals via the I²C bus
- **Light blue** highlights the LM75B temperature sensor

The layout of the elements on the bottom side of the Teensy Adaptor is provided next.



JUMPERS

All **Teensy Adaptor** jumpers are located on the bottom side:

1) **JP1, JP2 and JP3**. For selecting the address of the **LM75B** temperature sensor on the **I²C** bus. By default, the address is 1001111x (all jumpers are open). The table for selecting the address is provided below.

JP1 (A2)	JP2 (A1)	JP3 (A0)	LM75B address
CLOSE	CLOSE	CLOSE	1001000x
CLOSE	CLOSE	OPEN	1001001x
CLOSE	OPEN	CLOSE	1001010x
CLOSE	OPEN	OPEN	1001011x
OPEN	CLOSE	CLOSE	1001100x
OPEN	CLOSE	OPEN	1001101x
OPEN	OPEN	CLOSE	1001110x
OPEN	OPEN	OPEN	1001111x

2) **JP4 and JP6**. They are designed to disconnect the +5V power (jumper **JP4**) and +3V3 power (jumper **JP6**) from the **M1** slot with the **mikroBUS™** bus on the module. By default, power is connected (CLOSE).
3) **JP5 and JP7**. They are designed to disconnect the +5V power (jumper **JP5**) from the **Vin** pin and the +3V3 power (jumper **JP7**) from the **3V3** pins of the **TEENSY** slot. By default, power is connected (CLOSE).
4) **JP8 and JP9**. They are designed to disconnect the **SCL** (jumper **JP8**) and **SDA** (jumper **JP9**) signals of the **M1** slot from the common **I²C** bus. By default, the signals are connected (CLOSE).
5) **JP10 and JP11**. They select the **CSM** chip select signal of the **MBL** connector: **CS** (jumper **JP10**, pin 10 of the **Teensy** module) or **CS1** (jumper **JP11**, pin 2 of the **Teensy** module). By default, JP10 is OPEN and JP11 is CLOSE (**CS1** is selected). Only one of these jumpers can be closed at a time.
6) **JP12 and JP13**. For connecting 4.7 kOhm pull-up resistors to the **SDA** (jumper **JP12**) and **SCL** (jumper **JP13**) signals of the **I²C** bus of the **Teensy Adaptor**. By default, they are CLOSE, meaning the resistors are connected.

Summary of the default jumper states:

Jumper	Function	Default
JP1-JP3	LM75B I²C address (A2, A1, A0)	OPEN
JP4	+5V to M1 (mikroBUS™)	CLOSE
JP5	+5V to Vin pin of TEENSY slot	CLOSE
JP6	+3V3 to M1 (mikroBUS™)	CLOSE
JP7	+3V3 to 3V3 pins of TEENSY slot	CLOSE
JP8	SCL of M1 to common I²C bus	CLOSE
JP9	SDA of M1 to common I²C bus	CLOSE
JP10	CS (pin 10) to CSM of MBL	OPEN
JP11	CS1 (pin 2) to CSM of MBL	CLOSE
JP12	SDA pull-up 4.7 kOhm	CLOSE
JP13	SCL pull-up 4.7 kOhm	CLOSE

MEZZANINES

Any mezzanine from the **IoTextra** series listed on the website **http://www.makethingshappy.io** can be used as a mezzanine. The mezzanine is connected to the **HOST** connector of the **Teensy Adaptor** via a 12-pin cable (see **QUICK START**).

ACCESSORIES

The following accessories may be required to use the module:

- Mating (plug) two-pin connector with a 3.5mm pitch for module power
- Two-pin cables with JST PH connectors for the **VIN**, **5V-1** and **5V-2** connectors
- **I^2C** cable with a four-pin JST PH connector
- 12-pin cable (2×6, 2.54mm pitch) for connecting the **Teensy Adaptor** to **IoTextra** series mezzanines
- Two 8-pin headers with a 2.54mm pitch for the **MBL** and **MBR** connectors